ISD Projects

Interactive Systems Design

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Project Work

- Develop and evaluate a novel efficient text entry method for smart watches
- Steps:
  1. Search related literature
  2. Develop the technique
  3. Design the Experiment
  4. Develop the experimental software
  5. Analyze results
  6. Write the report

BubbleBoard

- Method based on a soft keyboard with a QWERTY layout (by V. Fuccella and B. Martin)
- Problem: the keyboard is too small to contain a full layout
- Solution: at the same time…
  - Only characters from a small set (bubbles) are enlarged and can be entered through a *tap*
  - Other characters cannot be pressed. They can become bubbles by pressing near their location
  - Il *Leipzig Corpora News-typical 2016* è un insieme di 1 milione di frasi scaricabile da http://wortschatz.uni-leipzig.de/en/download/

Language-based improvement

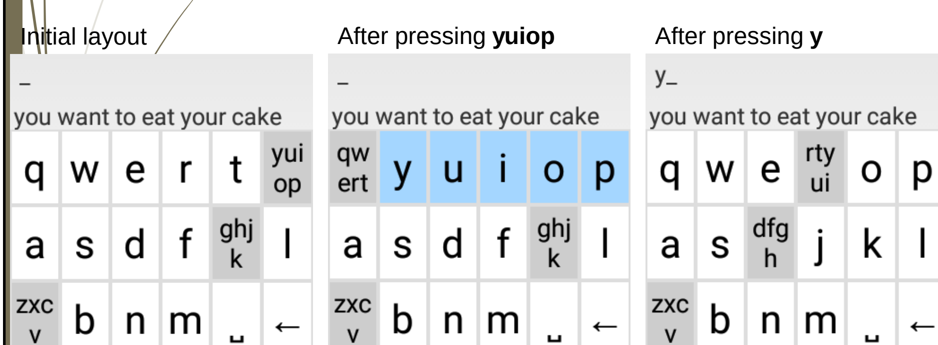
- In the worst case, the KSPC (number of taps to enter a character) of Bubbleboard is 2: one to zoom and one to select the character
- Improverment I: enlarge the most frequent characters (static layout)
- Improvement II: choice of the key to enlarge (dynamic layout)
  - After a character is entered, we can predict next and enlarge the corresponding key
  - E.g.: in Italian, after entering «q», the next character is «u» with very high probability
- Possible language model: «trigraph model»
  - I observe the last two character entered and predict the third one.
  - I need a corpus for language analysis: *Leipzig Corpora News-typical 2016* is a set of1 million phrases http://wortschatz.uni-leipzig.de/en/download/

BubbleBoard: static layout

- Best layout with 6 keys per line, max 3 char per key
- Estimated KSPC: 1.3257

BubbleBoard: dynamic layout

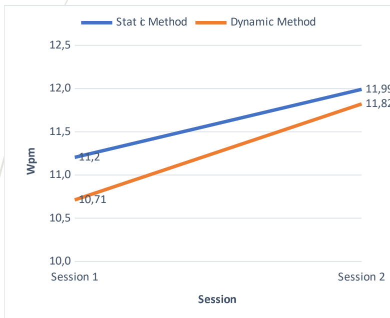
- Best layout with 6 keys per line
- Estimated KSPC: 1.1475



Preliminary Experiments

- Static VS Dynamic - 12 participants, 2 sessions of 5 ph

Speed



AVG SUS scores:
- Static: 81.45
- Dynamic: 75.83

Preference:
- Static: 25%
- Dynamic: 75%

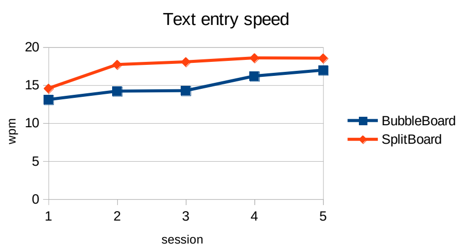
SplitBoard

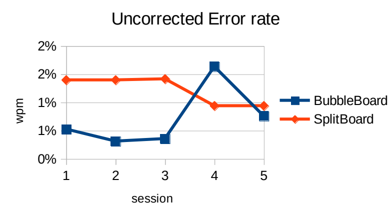
Hong, Jonggi, et al. "SplitBoard: A simple split soft keyboard for wristwatch-sized touch screens." Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems. ACM, 2015.

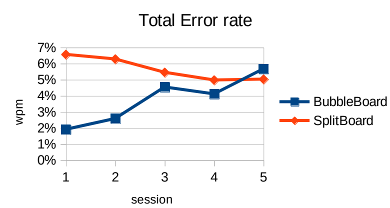
- Soft keyboard designed for smartwatches
- As the user flicks left or right on the keyboard, it switches between the left and right halves of a QWERTY keyboad.

Preliminary Experiment II

- **Static BubbleBoard VS SplitBoard**
- 2 participants, 5 sessions of 5 ph each







Project objectives

- Improve the implementation of BubbleBoard
- Perform experiments:
  - Compare versions
    - static, dynamic
    - Display shape (square, circular)
    - Ambiguous version?
  - Compare to other methods (SplitBoard, …)